

Research Paper: Deep Residual Networks for Telemetry Processing

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Abstract

This research paper investigates modern deep learning architectures for real-time telemetry processing in complex industrial environments. The paper evaluates residual networks and compares them with traditional recurrent neural networks.

1. Main Objective

The main objective of this paper is to design a high-throughput, low-latency deep residual neural network architecture that achieves robust signal filtering and feature extraction on edge computing hardware. The goal is to enable real-time processing under severe bandwidth and computational constraints.

2. Methodology

We implemented a 50-layer 1D residual convolutional neural network (ResNet-Signal) integrated with an adaptive Kalman filter for pre-processing. Training was conducted using 250,000 multi-channel telemetry samples with Adam optimizer, batch size 64, and cosine learning rate decay over 100 epochs.

3. Key Findings & Results

The main findings demonstrate that the proposed architecture achieves a 24.3 percent reduction in root mean square error (RMSE) compared to baseline LSTM networks, while reducing inference latency to 4.2 milliseconds per telemetry frame on standard edge devices.

4. Conclusion

In conclusion, residual networks provide significant efficiency gains for high-throughput sensor streams.